

Writing Piecewise Rules for Rates and Fees

Answer Key

Algebra 1

Quick Answers

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| 1. (b) the fare for 4 miles = 11, —
2. $C(2) = 7$, $C(9) = 19$
3. (b) the cost of a 5-hour visit = 14, —
4. (b) the charge for 8 gigabytes = 55, —
5. (a) the distance = 13, —
6. (b) the fine after 10 days = 1.75, — | 7. $E(250) = 30$, $E(500) = 72$
8. (b) the pay for a 46-hour week = 882, —
9. (a) the hours worked = 49, —
10. (b) the break-even number of minutes = 800, — |
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What is verified

2 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 3, 4, 5, 6, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-CED.A.1 — *problems 5, 9, 10*

HSA-CED.A.2 — *problems 1, 3, 4, 6, 8, 10*

HSF-IF.A — *problems 2, 7*

Difficulty 1–5 of 5 across 21 tagged checks.

- 1.** Taxi: \$3 pick-up plus \$2 per mile up to 5 miles; beyond that, \$13 plus \$1.50 per extra mile. (a) Write $C(m)$. (b) The fare for 4 miles.

(a) — instructor-judged. The rule is

$$C(m) = \begin{cases} 3 + 2m, & 0 \leq m \leq 5 \\ 13 + 1.5(m - 5), & m > 5 \end{cases}$$

Full credit needs the conditions written down and, in the second branch, both the 13 already charged at 5 miles and the $(m - 5)$ so the cheaper rate applies only to the miles past the fifth. A second branch written $1.5m$ is the standard error: it forgets everything already paid.

(b) 4 miles satisfies $0 \leq m \leq 5$, so use the first branch: $3 + 2(4) = 3 + 8$.

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- 2.** Using the taxi rule, find $C(2)$ and $C(9)$.

Solution: Check the condition before computing. $2 \leq 5$, so the first branch; $9 > 5$, so the second.

$$C(2) = 3 + 2(2) = 3 + 4$$

$$C(9) = 13 + 1.5(9 - 5) = 13 + 6$$

$$\boxed{C(2) = 7, \quad C(9) = 19}$$

3. Pool: \$5 covers up to 2 hours, then \$3 per extra hour. (a) Write $C(t)$. (b) A 5-hour visit.

(a) — **instructor-judged.** The rule is

$$C(t) = \begin{cases} 5, & 0 \leq t \leq 2 \\ 5 + 3(t - 2), & t > 2 \end{cases}$$

Full credit needs a first branch with no t in it at all — the price does not depend on the length of a short visit — and the $(t - 2)$ in the second, so the extra charge begins with the third hour.

(b) $5 > 2$, so the second branch: $5 + 3(5 - 2) = 5 + 9$.

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4. Phone plan: \$25 up to 5 GB, then \$10 per extra GB. (a) Write $F(g)$. (b) The charge for 8 GB.

(a) — **instructor-judged.** The rule is

$$F(g) = \begin{cases} 25, & 0 \leq g \leq 5 \\ 25 + 10(g - 5), & g > 5 \end{cases}$$

Both conditions must be stated, and the extra \$10 must apply only beyond the fifth gigabyte.

(b) $8 > 5$, so $25 + 10(8 - 5) = 25 + 30$.

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5. A taxi ride cost \$25. (a) How far was it? (b) How did you know which branch to use?

(b) — **instructor-judged.** At exactly 5 miles the fare is $3 + 2(5) = \$13$, the largest fare the first branch can ever produce. Since \$25 is more than that, the ride must be longer than 5 miles, so the second branch applies. Full credit requires that comparison with the boundary value *and* a check at the end that the answer really is more than 5. Trying a branch at random is not the method.

Solution (a):

$$13 + 1.5(m - 5) = 25$$

$$1.5(m - 5) = 12$$

$$m - 5 = 8$$

The ride was 13 miles, and $13 > 5$, which confirms the branch was right.

13

6. Library: no fine for 3 days, then \$0.25 per day beyond the third. (a) Write $F(d)$. (b) The fine at 10 days.

(a) — **instructor-judged.** The rule is

$$F(d) = \begin{cases} 0, & 0 \leq d \leq 3 \\ 0.25(d - 3), & d > 3 \end{cases}$$

Full credit needs the exact zero on the first branch and the $(d - 3)$ on the second, so that a book four days late costs 25 cents rather than a dollar.

(b) $10 > 3$, so $0.25(10 - 3) = 0.25 \times 7$.

1.75

7. Electricity: \$0.12 per kWh up to 300 kWh, then \$36 plus \$0.18 per extra kWh. Find $E(250)$ and $E(500)$.

Solution: $250 \leq 300$ uses the first branch; $500 > 300$ uses the second.

$$E(250) = 0.12(250)$$

$$E(500) = 36 + 0.18(500 - 300) = 36 + 36$$

$$E(250) = 30, \quad E(500) = 72$$

Note that the two branches agree at $k = 300$: $0.12(300) = 36$. A price list that jumped at the boundary would charge less for more electricity.

8. Pay: \$18/hour up to 40 hours, \$27/hour beyond. (a) Write $P(h)$. (b) Pay for 46 hours.
(a) — **instructor-judged.** The rule is

$$P(h) = \begin{cases} 18h, & 0 \leq h \leq 40 \\ 720 + 27(h - 40), & h > 40 \end{cases}$$

The 720 is 18×40 , the pay already earned at the boundary; the overtime rate applies only to $(h - 40)$. Writing the second branch as $27h$ pays overtime on the whole week and is the classic error.

(b) $46 > 40$, so $720 + 27(46 - 40) = 720 + 162$.

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9. A worker earned \$963 in a week. (a) How many hours? (b) How did the amount tell you the branch?

(b) — **instructor-judged.** A full 40-hour week pays $18 \times 40 = \$720$, which is the most the first branch can produce. \$963 exceeds it, so the week ran into overtime. Full credit requires that comparison and a final check that the answer really exceeds 40.

Solution (a):

$$720 + 27(h - 40) = 963$$

$$27(h - 40) = 243$$

$$h - 40 = 9$$

The worker put in 49 hours, and $49 > 40$, confirming the branch.

49

10. **Synthesis challenge.** Plan A: $A(n) = 30 + 0.1n$. Plan B: \$20 for the first 200 minutes, then \$0.15 per extra minute. (a) Write $B(n)$. (b) Where do they cost the same? (c) Which is cheaper below that?

(a) — **instructor-judged.** The rule is

$$B(n) = \begin{cases} 20, & 0 \leq n \leq 200 \\ 20 + 0.15(n - 200), & n > 200 \end{cases}$$

Both conditions and the $(n - 200)$ are required.

Solution (b): Set the two charges equal on the region where $n > 200$.

$$30 + 0.1n = 20 + 0.15(n - 200)$$

$$30 + 0.1n = 0.15n - 10$$

$$40 = 0.05n$$

So the plans cost the same at $n = 800$ minutes, and $800 > 200$, so the branch used was the right one. $n = 800$

(c) — instructor-judged. Plan B is cheaper for every number of minutes below 800 and dearer above it. Full credit requires evidence, not an assertion: either a test value on each side — at $n = 400$, $A = \$70$ and $B = \$50$; at $n = 1000$, $A = \$130$ and $B = \$140$ — or the rate argument, that past 200 minutes B climbs \$0.15 a minute while A climbs only \$0.10, so B must eventually overtake A, and 800 is where it does.